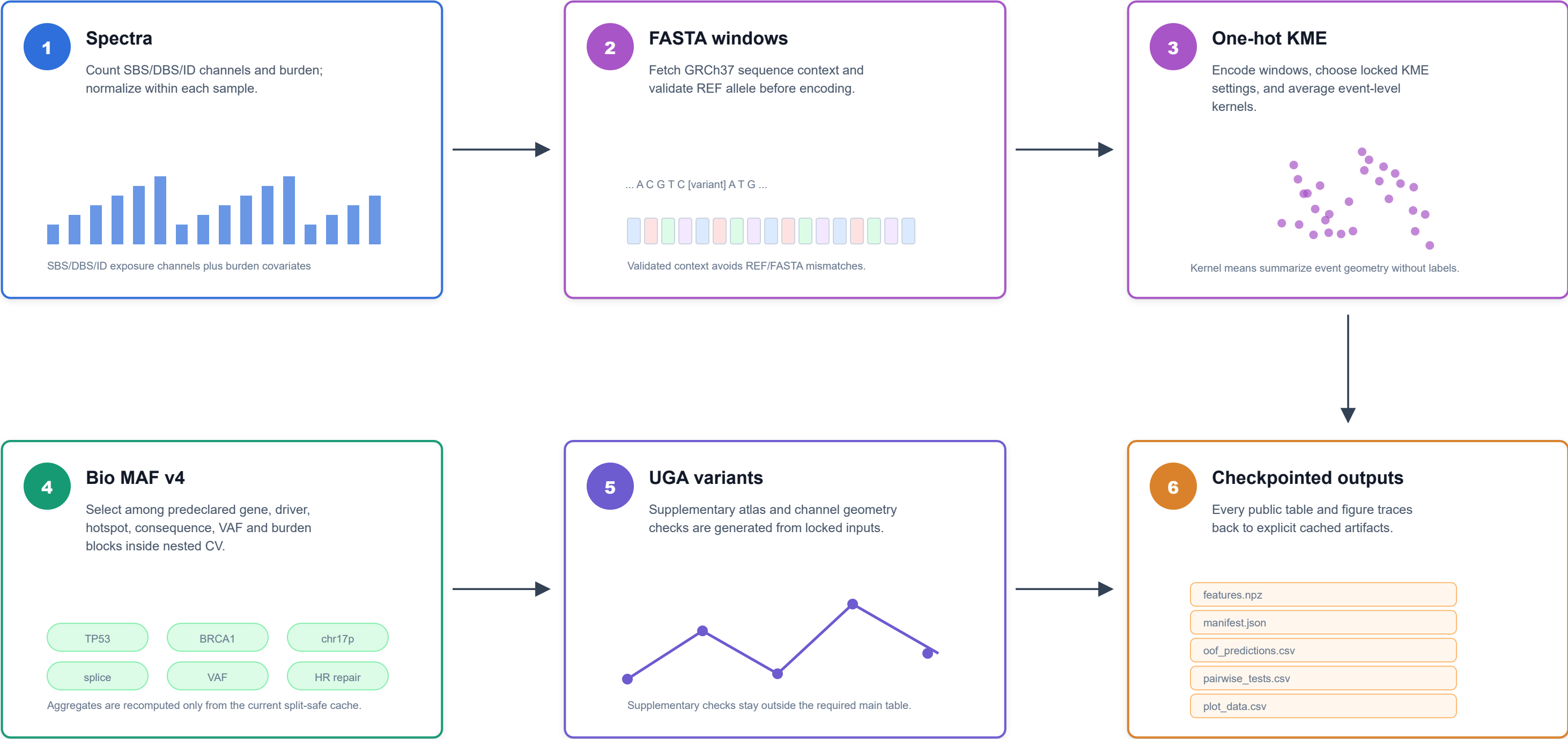


Supplementary Figure S1. Representation construction

Each feature block is cached with input fingerprints, split provenance, FASTA provenance, parameters and sample manifests.

Cards show the reproducible construction path used for strict no-leakage manuscript outputs.



Strict reruns invalidate only stale feature caches; checkpointed OOF predictions and tests preserve reproducibility without rerunning unaffected slots.

All artifact names are drawn inside the card bounds to mirror the HTML and Word export layout.